

HEEKUN ROH

hroh@mit.edu

Ph.D. Candidate at Massachusetts Institute of Technology, United States

EDUCATION

Massachusetts Institute of Technology (MIT) Sep. 2023 - Present

Ph.D. Candidate in Aeronautics and Astronautics

Advisor : Prof. Olivier L. de Weck (Engineering Systems Laboratory)

Keywords : Systems Engineering, Complex Systems, Space Systems, Space Situational Awareness

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2017 - Feb. 2019

M. S. in Aerospace Engineering

Advisor : Prof. Min-Jea Tahk

Keywords : Guidance and Control, Trajectory Optimization, Optimal Resource Allocation and Planning

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2013 - Feb. 2017

B.S. in Aerospace Engineering and Electrical Engineering

Summa Cum Laude, KAIST Presidential Fellow

PROFESSIONAL EXPERIENCE

Satrec Initiative Feb. 2019 - Oct. 2022

Associate Systems Engineer

Keywords : Satellite Attitude Determination and Control, Onboard Software Development and Testing, Mission Analysis, Verification and Validation, Commissioning

RESEARCH EXPERIENCE

Massachusetts Institute of Technology (MIT) Sep. 2023 - Present

Strategic Engineering Research Group (SERG)

Graduate Research Assistant

KAIST, Flight Dynamics and Control Lab (FDCL) Dec. 2022 - Aug. 2023

Department of Aerospace Engineering, Appointed Researcher

KAIST, Flight Dynamics and Control Lab (FDCL) Dec. 2016 - Feb. 2019

Department of Aerospace Engineering, Master's Student

PUBLICATIONS

Journal Papers

- [1] H. Roh, Y.J. Oh, M.J. Tahk, K.J. Kwon, and H.H. Kwon, "L1 Penalized Sequential Convex Programming for Fast Trajectory Optimization: With Application to Optimal Missile Guidance," *International Journal of Aeronautical and Space Sciences (IJASS)*, Vol. 21, pp. 493-503, Jun. 2020.
- [2] H. Roh, Y.J. Oh, M.J. Tahk, and Y.R. Jung "Optimal Weapon-Target Assignment of Multiple Dissimilar Closed-In Weapon Systems Using Mixed Integer Linear Programming," *Journal of Korean Society for Aeronautical and Space Sciences*, Vol. 47, No. 11, pp.787-794, Nov. 2019.
- [3] H. Roh, S.W. Shim, and M.J. Tahk, "Maneuver Algorithm for Bearings-Only Target Tracking with Acceleration and Field of View Constraints," *International Journal of Aeronautical and Space Sciences (IJASS)*, Vol. 19, No. 2, pp. 423-432, Jun. 2018.

- [4] S.Y. Han, J.H. Bai, S.M. Hong, H. Roh, M.J. Tahk, J.S. Yun, S.H. Park, “Control law for agile turn of air-to- air missile during boost phase,” *International Journal of Aeronautical and Space Sciences(IJASS)*, Vol.18, No.4, Dec. 2017

International Conference Papers

- [1] H. Roh, L. Etzenbach, A. Oltramare, J. Norheim, O.L. De Weck. “Factored Dependency Structure Matrix for Representation of Multi-Connection Systems,” 27th International Dependency and Structure Modeling Conference (DSM 2025), Hoboken, NJ, USA, Sep. 2025.
- [2] H. Roh, L. Etzenbach, A. Oltramare, J. Norheim, O.L. De Weck. “Size Constrained K-Means Clustering for Controlled Design Structure Matrix Partitioning,” 2025 IEEE International systems Conference (SysCon), Montreal, Canada, Apr. 2025.
- [3] L. Etzenbach, H. Roh, A. Oltramare, J. Norheim, O.L. De Weck. “The Road Ahead: Present and Future Trends in Global Civil Aircraft Regulations,” 2025 IEEE International systems Conference (SysCon), Montreal, Canada, Apr. 2025.
- [4] L. Etzenbach, H. Roh, A. Oltramare, J. Norheim, O.L. De Weck. “Design-Driven Framework for the Complexity Quantification of Mechanical Components,” 2025 IEEE International systems Conference (SysCon), Montreal, Canada, Apr. 2025.
- [5] H. Roh, and O.L. de Weck, “Global Visibility Assessment Method of In-Situ Optics Constellation for Uncatalogued Resident Space Objects,” *AIAA Scitech Forum 2025*, Orlando, Florida, USA, Jan. 2025.
- [6] M.J. Tahk, J.C. Park, H. Roh, C.H. Lee., “Suboptimal guidance based on pursuit and impact angle control for long-range air-to-air missiles,” 14th International Conference on Mechanical and Aerospace Engineering (ICMAE), Porto, Portugal, Jul. 2023
- [7] J.C. Park, H. Roh, C.H. Lee, M.J. Tahk, H.S. Choi, S.Y. Kim. “Nonlinear Three-Loop Autopilot Design for Hypersonic Vehicles Considering Aerodynamic-Propulsion Dynamics Couplings,” 14th International Conference on Mechanical and Aerospace Engineering (ICMAE), Porto, Portugal, Jul. 2023
- [8] H. Roh, T. Kim, Y. Moon, H. Woo, C.H. Lee, “Optimal Time Terrestrial Mapping Scheduling for Spaceborne Push-broom Imagers,” Aerospace Europe Conference 2023 (EUCASS-CEAS 2023), Lausanne, Switzerland, Jul. 2023.
- [9] Y.J. Oh, H. Roh, and M.J. Tahk, “Fast Trajectory Optimization using Sequential Convex Programming with No-Fly Zone Constraints,” 21st IFAC Symposium on Automatic Control in Aerospace (ACA), Cranfield, United Kingdom, Aug. 2019.
- [10] H. Roh, Y.J. Oh, M.J. Tahk, and C.H. Lee, “Fast Trajectory Optimization Using Sequential Convex Method for Guided Missiles,” *The 5th CEAS Conference on Guidance, Navigation and Control (EuroGNC)*, Milano, Italy, Apr. 2019.
- [11] Y.J. Oh, H. Roh, M.J. Tahk, “A Lattice Path Following Algorithm for Guided Missiles,” *2018 Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, Chengdu, China, Oct. 2018.
- [12] H. Roh, M.H. Cho and M.J.Tahk, “Trajectory Optimization Using Cramer-Rao Lower Bound for Bearings-Only Target Tracking,” *AIAA Scitech Forum 2018*, Kissimmee, Florida, USA, Jan. 2018.
- [13] H. Roh, B.Y. Lee, and M.J. Tahk, “Automatic Maneuver Generation for Suppression of Enemy Air Defense Using Scoring Function Matrix,” *Asia-Pacific International Symposium on Aerospace Technology (APISAT)*, pp.1087-1093, Seoul, Korea, Oct. 2017.

- [14] J.M. Park, S.M. Hong, H. Roh, M.J. Tahk, Y.Y. Kim, J.S. Yun, "Optimal control of roll-pitch seeker with singularity avoidance," *The 26th Mediterranean Conference on Control and Automation (MED)*, Zadar, Croatia, Jun. 2018.
- [15] J.M. Park, H. Roh, M.J. Tahk, "Co-evolutionary Method For Dynamic Weapon-Target Assignment," *Advances in Control and Optimization of Dynamic Systems(ACODS)*, Hyderabad, India, Feb. 2018.
- [16] S.Y. Han, J.H. Bai, H. Roh, S.M. Hong, M.J. Tahk, J.S. Yun, S.H. Park, "Three-Dimensional Velocity Maximizing Agile Turn of Air-to-Air Missile with Collision Triangle Constraint," *25th Mediterranean Conference on Control and Automation (MED)*, Valletta, Malta, Jul. 2017.

Domestic Conference Papers

- [1] H. Roh, M.J. Tahk, K.B. Kim, and H.H. Kwon, "Trajectory Optimization for Missile Impact Time Control Problem Using L1 Penalty Method and Sequential Convex Programming," *The Korean Society for Aeronautical and Space Sciences(KSAS): 2018 Fall Conference*, Jeju, Korea, Nov. 2018.
- [2] H. Roh, and M.J. Tahk, "Comparison Study on Bearings-Only Target Tracking Filters," *The Society for Aerospace System Engineering(SASE): 2018 Fall Conference*, Gyeongju, Korea, Nov. 2018.
- [3] H. Roh, S.M. Hong, M.J. Tahk, K.B. Kim, K.J. Kwon, and H.H. Kwon, "Optimal Impact Time Control Guidance Using Convex Optimization ," *Korean Institute of Military Science and Technology(KIMST): 2018 General Conference*, Jeju, Korea, Jun. 2018.
- [4] H. Roh, and M.J. Tahk, "Optimization of Closed-In Weapon System Target Assignment Using Mixed Integer Linear Programming," *The Korean Society for Aeronautical and Space Sciences(KSAS): 2018 Spring Conference*, Gyeongju, Korea, Apr. 2018.
- [5] H. Roh, B.Y. Lee, and M.J. Tahk, "Maneuver Generation for Moving Obstacle Avoidance Using Scoring Function Matrix," *The Korean Society for Aeronautical and Space Sciences(KSAS): 2017 Fall Conference*, Jeju, Korea, pp. 561-562, Nov. 2017.
- [6] H. Roh, J.M. Park, and M.J. Tahk, "Modeling and Formulation for Return-to-Launch-Site Trajectory Optimization of Reusable Launch Vehicle," *The 17th Symposium on Space Launch Vehicle Technology*, Gyeongju, Korea, Aug. 2017
- [7] K. Kim, H. Kim, H. Roh, and H.L. Choi, "Flying BioLab : A CanSat platform for sampling and monitoring air bacteria in bio-hazardous area" *The Korean Society for Aeronautical and Space Sciences(KSAS): 2014 Fall Conference*, Jeju, Korea, Nov. 2014.

HONOR & AWARDS

Graduation with Highest Honors, Summa Cum Laude	Feb. 2017
KAIST Presidential Fellowship	Mar. 2013 - Feb. 2017
Selected 20 students who show academic excellence and leadership	
Boeing Scholarship	Mar. 2014 - Dec. 2016
Selected students in Department of Aerospace Engineering, KAIST	
Dean's List, College of Engineering	Mar. 2015
The World Embedded Software Contest 2014, 1st Place	Dec. 2014
in 'Medical Services' section	
Organized by Ministry of Trade, Industry and Energy	

Samsung SDS Software Club Championship 2014, 1st Place Organized by Samsung SDS	Nov. 2014
Cansat Competition Korea 2014, 2nd Place Organized by KAIST Satellite Technology Research Center (SATREC)	Aug. 2014
Cansat Competition Korea 2012, 1st Place Organized by KAIST Satellite Technology Research Center (SATREC)	Aug. 2012

SERVICES

Guest Lectures

- Tecnológico de Monterrey, A Brief Introduction to Space Systems Engineering (2026)

Professional Memberships

- The International Council on Systems Engineering (INCOSE)
- The American Institute of Aeronautics and Astronautics (AIAA)
- The Korean Society for Aeronautical and Space Sciences (KSAS)

Reviewer Activities

- IEEE Transactions on Aerospace and Electronic Systems (2022, 2024)
- AIAA Journal of Spacecraft and Rockets (2024)
- International Journal of Aeronautical and Space Sciences (2025, 2026)
- (Conference) Complex Adaptive Systems (2026)

Open Source Contributions

- C89 Implementation of the MINRES Algorithm (2023)
<https://web.stanford.edu/group/SOL/software/minres/>

LANGUAGE PROFICIENCY

Korean

Native

English

Full Professional Proficiency

TOEFL: (110, R30/L30/S25/W25, 2022)

GRE: (V165/Q170/AW 5.0, 2022)

French

Advanced Working Proficiency

DELF B2 (2016)

TECHNICAL SKILLS

Programming Languages	Fluent in C, Python, MATLAB Experiences with Java, Julia, and many others
Optimization Softwares	Fluent in MOSEK, GPOPS II, Gurobi Developed in-house QP, SOCP interior point solvers in C, MATLAB
Miscellaneous	L ^A T _E X, Atlassian SWs, STK